MIL-C-60937 (MU)
30 November 1967

MILITARY SPECIFICATION

CONTAINER, AMMUNITION, METAL, PA 20:
FOR
30MM CARTRIDGES

1 SCOPE

1 1 This specification covers the PA 20 Ammunition container for packing 30MM Cartridges

2. APPLICABLE DOCUMENTS

2 1 The following documents of the issue in effect on date of invitation for bids or request for proposal. form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-490	- Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-I-45607	- Inspection Equipment, Supply and Maintenance of

STANDARDS

FEDERAL

FED-STD-151	- Metals; Test Methods

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-171	- Systems for Preparation Painting and Finishing Metal and Wood Surfaces

FSC: 8140

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

MIL-STD-414	-	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. Army Munitions Command

9229066	-	Container, Ammunition, Metal, PA 20:

PUBLICATIONS

ARMY

ET-9229066	-	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts and assemblies.-The parts and assemblies shall comply with all requirements specified on Drawing (Dwg.) 9229066, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Airtightness.-The metal container shall withstand without leakage, an air pressure differential of three (3) pounds per square inch (p.s.i.) for fifteen (15) seconds minimum (min.) when tested as specified in 4.4.2.

3.4 Functioning.-The cover of the container shall open, close and be removed without binding or requiring exertion of undue force. Gaskets shall not stick to top edges of bodies, be cut or split nor have shifted within the gasket retainer when the cover is removed when tested as specified in 4.4.9.

3.5 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner. They shall be free of burrs, nicks, sharp edges, cracks, dirt, surface defects, grease, rust and other foreign matter.

3.6 Submission of results of contractor conducted examinations and tests. The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.4)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

3.7 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

4.2 First article inspection.

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1) The first article sample shall consist of the following:

Test	Sample
Gasket compression	Ten (10) Metal containers
Airtightness	Fifty (50) Metal containers
Salt Spray	Two (2) Body Assemblies and Cover assemblies, less gasket
Paint Adhesion	Two (2) Body assemblies and cover assemblies
Paint Thickness	Two (2) Body assemblies and cover assemblies
Phosphate weight	One (1) Test panel (4 hour maximum process interval)
Weld security	Twenty (20) Cover assemblies
Weld security	Twenty (20) Body assemblies
Functioning	Twenty (20) Metal container assemblies

All parts and completed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions.

4.3.1 Lot formation.-The term "Lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications, and same specification revisions. Changes to the process, specifications, or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105.

Unless otherwise approved by the Contracting Officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period, nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 (ABC-STD-105), except that, inspection for Critical defects shall be 100 percent. Contractor's sampling plans, if used, shall be approved by the Government, and shall provide, as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each Major defect and an AQL of 0.65 percent for each Minor defect except where 100% inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance 4.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

4.3.2.1 Cover assembly (See Dwg. 9230098 covering a detail of Dwg. 9230097)

Categories	Defects	Method of Inspection	Defect Code (See 6.2)
Critical:	None defined		
Major:	AQL 1.00 Percent		
101.	Concavity and convexity of width	Gage	01001
102.	Concavity and convexity of length	Gage	01002
103.	Gasket defective (See 6.6)	Visual	01003
104.	Gasket inverted (See 6.7)	Visual	01004
105.	Loose gasket (See 6.8)	Manual	01005
106.	Component missing or improperly positioned	Visual	01006
107.	Protective coating with bare spot	Visual	01007
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	01008

4.3.2.2 Body assembly (See Dwg. 9230095 covering a detail of Dwg. 9229066)

Categories	Defects	Method of Inspection	Defect Code (SEE 6.2)
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Concavity and convexity of length	Gage	02001
102.	Component missing or improperly positioned	Visual	02002
103.	Protective coating with bare spot	Visual	02003
104.	Concavity and convexity of width	Gage	02004
Minor:	AQL 1.50 Percent		
201.	Inside length, min	Scale	02005
202.	Inside width, min	Scale	02006
203.	Inside depth, min	Scale	02007
204.	Evidence of poor workmanship (See 3.5)	Visual	02008

4.3.2.3 Metal container assembly (See Dwg. 9229066)

Categories	Defects	Method of Inspection	Defect Code
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Marking misleading or unidentifiable	Visual	03001
Minor:	AQL 2.50 Percent		
201.	Outside height	Gage	03002
202.	Outside length	Gage	03003
203.	Outside width	Gage	03004
204.	Evidence of poor workmanship (See 3.5)	Visual	03005
205.	Patch of blue color missing (when applicable)	Visual	03006

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

4.3.3 Testing.-

4.3.3.1 Gasket compression (See Dwg. 9230097), Major defect, Code No. (04001).-The sampling plan for this test shall be in accordance with MIL-STD-414, Part II, Section B with an AQL of 1.50 Percent. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.2 Airtightness (See Paragraph 3.3), Major defect, Code No. (05001).-Twenty-five (25) metal containers shall be selected from each lot and subjected to this test. If two (2) or more containers fail to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.3.3.3 Salt Spray (See Dwgs. 9230095 and 9230098, less gasket), Major defect, Code No. (06001).-The sampling and rejection criteria shall be in accordance with TT-C-490. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4.

4.3.3.4 Paint adhesion (See Dwgs. 9230095 and 9230098), Major defect, Code No. (07001).-The sampling plan for this test shall be in accordance with MIL-STD-105, single plan, with an AQL of 0.40 Percent. The test shall be performed as specified in 4.4.4.

4.3.3.5 Paint thickness (See Dwgs. 9230095 and 9230098), Major defect, Code No. (08001).-The sampling plan for this test shall be in accordance with MIL-STD-105, single plan, with an AQL of 0.40 Percent. The test shall be performed as specified in 4.4.5 using equipment in accordance with 4.3.4.

4.3.3.6 Phosphate weight (See Dwg. 9230095 and 9230098), Major defect, Code No. (09001).-The sampling and rejection criteria shall be in accordance with TT-C-490. The test shall be performed as specified in 4.4.6.

4.3.3.7 Security of weld on cover assembly (See Dwg. 9230098), Major defect, Code No. (10001).-The sampling plan for this test shall be in accordance with MIL-STD-105, single plan, with an AQL of 0.40 Percent. The test shall be performed as specified in 4.4.7 using the equipment specified in 4.3.4.

4.3.3.8 Security of weld on body assembly (See Dwg. 9230095), Major defect, Code No. (11001).-The sampling plan for this test shall be in accordance with MIL-STD-105, single plan, with an AQL of 0.40 Percent. The test shall be performed as specified in 4.4.8 using equipment in accordance with 4.3.4.

4.3.3.9 Functioning (See Para. 3.4), Major defect, Code No. (12001).-The sampling plan for this test shall be in accordance with MIL-STD-105, single plan, with an AQL of 1.00 Percent. The test shall be performed as specified in 4.4.9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 9229066 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall, when required by the Government, make available all inspection and test equipment necessary for determining conformance with the contract requirements. Personnel for operating the equipment, and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 Gasket compression.-The gage shown on the Equipment List will be used to determine the gasket compression. The following procedure will be used:

a. With the box body resting on a surface plate or other suitable stabilized platform (See 6.5) place the box cover on the box body with the components in the unlatched position. Measure the height of box from the platform to the top of the cover in 4 places, 2 readings being taken on each side of the box at 4 inches from the ends of the box. Add the measurements and divide by 4. This is the average free height of the box.

b. Without moving the box from the location used to determine the free height, close the box by engaging the hasps and latches. Remeasure the box at the same positions measured to determine the free height of the box. Add these measurements and divide by 4. This is the average closed height of the box.

c. Subtract the average closed height of the container from the average free height of the container. The result will be considered the gasket compression. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.2 Airtightness.-The metal container shall be placed in the fixture shown on the Equipment List and a measured quantity of air shall be applied to produce the required air pressure. Observation shall be made for bubbles escaping from the interior of the container. This test is a non-destructive test. Parts so tested may be returned to the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

4.4.3 Salt Spray.-The parts selected shall be subjected to the salt spray test in accordance with Standard No. 151, Method 811.1 using a 5% sodium chloride solution. This is a destructive test. Parts so tested shall not be returned to the lot.

4.4.4 Paint adhesion.-The test will be conducted in accordance with Paragraph 6.7 MIL-STD-171. Three places randomly located on the surface of the sample component will be tested to assure adequate paint adhesion on all surfaces. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.5 Paint thickness.-The test shall be conducted in accordance with Paragraph 6.5 of MIL-STD-171.

4.4.7 Security of weld on cover assembly.-The cover assembly shall be placed in the fixture shown on the Equipment List and the pull specified on Dwg. 9230098 shall be applied. Observation shall be made for the requirement specified on Dwg. 9230098. Any damage or distortion other than weld separation resulting from this test will be disregarded. However, damaged or distorted assemblies will not be returned to the lot.

4.4.8 Security of weld on body assembly.-The body assembly shall be placed in the fixture shown on the Equipment List and the pull specified on Dwg. 9230095 shall be applied. Observation shall be made for the requirement specified on Dwg. 9230095. Any damage or distortion other than weld separation resulting from this test will be disregarded. However, damaged or distorted assemblies will not be returned to the lot.

4.4.9 Functioning.-The metal container assembly will be tested by opening and closing both latch assemblies to check functioning of hasp and latch assemblies and ease removal of cover. Also to assure that the gaskets are snug within the gasket retainer and that the gaskets do not adhere to the top edges of the bodies. This test is a non-destructive test. Parts so tested may be returned to the lot.

5. PREPARATION FOR DELIVERY

5.1 Packing.-

5.1.1 Level C.-Pack individually in accordance with Para. 3.3.1.2 of MIL-STD-1169.

5.2 Marking.-Marking of the carton or boxes shall be in accordance with MIL-STD-1169.

5.3 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:
a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of First Article Samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.6.

6.3.1 Equipment Lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings of Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.6. Action required by the contractor with respect to the use of commercial equipment is detailed in 6.3.5.

6.3.2 Equipment lists codes.-The inspection equipment as defined in 6.3.3, 6.3.4 and 6.3.5 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army designs, the contractor shall comply with, and use these designs, and with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (See 6.3.4) or commercial equipment selections (See 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes, and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.6.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors, normally, are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.4 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.6.

6.3.6 Submission of contractor designs.-All submitted designs shall reference the applicable five-digit code number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND1C. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.

6.5 Three point contact with the box resting on blocks may be used to overcome any rocking motion which might be encountered with the box resting on a flat surface. However, since the measurement technique relies upon relative distances, care should be taken that there is no displacement on the reference surface between the readings taken in the open and closed positions.

6.6 Gasket which is split, cut, pitted, torn, cracked or which shows porosity or flash on the surface intended to mate with the body, or which contains weak sections or bubbles discernible in the assembly.

6.7 Gasket assembled with the relief groove showing.

6.8 Gasket which has side play or end play in the cover assembly. Finger pressure or shaking of the cover will determine gasket looseness.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 8140-A151

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:06Z
Check the source to verify that this is the current version before use.